Group and Representation Theory

Chapter 3. $SU(2)$

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J_3 Eigenstates

- $SU(2)$ algebra :

$$[J_i, J_j] = i\epsilon_{ijk} J_k \tag{1}$$

This is the simplest (nonabelian) compact Lie algebra, since there is no simpler totally antisymmetric object than ϵ_{ijk}

- Since no two generators commute, we can diagonalize only one of them at a time, say J_3 :

$$J_3 \left|j, \alpha\right\rangle = j \left|j, \alpha\right\rangle \tag{2}$$

j is the largest eigenvalue of J_3, called the **highest weight**, and α labels possibly degenerate eigenstates with the same eigenvalue j

J_3 Eigenstates

- From now on we consider a **finite** vector space, which transforms under an irrep of the $SU(2)$ algebra
- The procedure is familiar from quantum mechanics, but only difference is that we do not use Casimir J^2

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Raising/Lowering Operators

We define **raising/lowering operators** $J^{\pm}$ by

$$J^{\pm} \equiv \frac{1}{\sqrt{2}} \left(J_1 \pm i J_2 \right) \tag{3}$$

From $SU(2)$ algebra, one can immediately check that

$$[J_3, J^{\pm}] = \pm J^{\pm} \tag{4}$$

$$[J^{+}, J^{-}] = J_3 \tag{5}$$

and

$$J_3 J^{\pm} |m\rangle = (m \pm 1) J^{\pm} |m\rangle \tag{6}$$

Therefore $J^{\pm} |m\rangle$ is an eigenstate of J_3, with raised/lowered eigenvalue $m \pm 1$!

Raising/Lowering Operators

Since j is the highest weight,

$$J^+ |j, \alpha\rangle = 0 \quad \forall \alpha \tag{7}$$

now $J^- |j, \alpha\rangle$ is an eigenstate of J_3 with the eigenvalue $j - 1$, so we write the proportionality constant $N_j(\alpha)$

$$J^- |j, \alpha\rangle = N_j(\alpha) |j - 1, \alpha\rangle \tag{8}$$

Since $\langle j, \alpha | j, \beta\rangle = \delta_{ab}$,

$$\langle j, \beta | J^+ J^- |j, \alpha\rangle = \langle j, \beta | (J^- J^+ + J^3) |j, \alpha\rangle \tag{9}$$

$$= N_j(\beta)^* N_j(\alpha) \tag{10}$$

$$= j\delta_{\alpha\beta} \tag{11}$$

Raising/Lowering Operators

Therefore, we conclude

$$J^+ |j-1, \alpha\rangle = \sqrt{j}\, |j, \alpha\rangle \tag{12}$$

$$J^- |j, \alpha\rangle = \sqrt{j}\, |j-1, \alpha\rangle \tag{13}$$

We obtain the recursion relation

$$N^2_{j-k} = \langle j-k, \alpha | J^+ J^- |j-k, \alpha\rangle \tag{14}$$

$$= \langle j-k, \alpha | J_3 + J^- J^+ |j-k, \alpha\rangle \tag{15}$$

$$= j - k + N^2_{j-k+1} \tag{16}$$

Raising/Lowering Operators

$$\begin{aligned} N_j^2 &= j \\ N_{j-1}^2 - N_j^2 &= j-1 \\ &\vdots \\ N_{j-k}^2 - N_{j-k+1}^2 &= j-k \end{aligned}$$

Therefore, we conclude

$$N_{j-k} = \sqrt{\frac{1}{2}(2j-k)(k+1)} \tag{17}$$

Raising/Lowering Operators

Or letting $m = j - k$,

$$N_m = \sqrt{\frac{1}{2}(j+m)(j-m+1)} \tag{18}$$

We've assumed that the vector space is finite-dimensional, so the lowering process can't continue indefinitely

$$J^- \left|j - x, \alpha\right\rangle = 0 \tag{19}$$

for some x, but this implies

$$N_{j-x}^2 = \frac{1}{2}(2j - x)(x + 1) = 0 \tag{20}$$

so $x = 2j$, and the **lowest state** is $\left|-j, \alpha\right\rangle \to 2j + 1$ states!

Raising/Lowering Operators

Implications :

- Since x is a nonnegative integer, j can be a half-integer
- Now the vector space breaks up into subspaces that are **invariant** under the algebra, since J_1 and J_2 can be expressed in terms of $J^{\pm}$
- But we've assumed that our vector space is an irrep, so we can neglect α!

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The Standard Notation

Now we denote the **highest weight** or **spin** by j and J_3 value as m

$$|j, m\rangle \tag{21}$$

States $|j, m\rangle$ transform under the **spin-j** irrep of $SU(2)$.

In the spin-j irrep, the matrix elements of J_3 and $J^{\pm}$ are

$$[J_3]_{m'm} = m\delta_{m'm} \tag{22}$$

$$[J^{\pm}]_{m'm} = \sqrt{\frac{1}{2}(j \mp m)(j \pm m + 1)}\delta_{m'm\pm1} \tag{23}$$

Example - Spin Half

The spin-half representation is given by

$$J_i = \frac{\sigma^i}{2}, (i = 1, 2, 3) \tag{24}$$

where σ^i are **Pauli matrices**
Finite rotations in spin-half rep are obtained by exponentiation of the generators J_i

$$\mathbf{R} = e^{i\vec{\alpha}\cdot\vec{\sigma}/2} \tag{25}$$

Example - Spin Half

For example, rotation by θ about z-axis would be

$$R_z(\theta) = \exp\left(i\theta\frac{\sigma^3}{2}\right) \tag{26}$$

from the Pauli matrix identity

$$e^{i\theta\sigma^k} = I_2\cos\theta + i\sin\theta\sigma^k \tag{27}$$

$$R_z(\theta) = \begin{pmatrix} \cos\frac{\theta}{2} + i\sin\frac{\theta}{2} & 0 \\ 0 & \cos\frac{\theta}{2} - i\sin\frac{\theta}{2} \end{pmatrix} \tag{28}$$

Example - Spin Half

Let χ be a **spinor**, and

$$R_z(2\pi)\chi = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \chi = -\chi \tag{29}$$

Surprisingly, a spinor gets an extra minus factor under 2π-rotation, and returns to itself after 4π rotation!



Figure 1: $\cdots$.

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Tensor Products

Now we've found spin-j irreps of $SU(2)$:

$$0, 1/2, 1, 3/2, 2, \cdots \tag{30}$$

Now we consider decomposition of tensor products into irreps, or the **angular momentum addition**

The idea is still the same - starting from the highest weight state, apply lowering operators

Tensor Products

Quick review on tensor products :

1. We usually denote $|i\rangle \otimes |x\rangle = |i, x\rangle$, where $|i\rangle$ transforms under the irrep D_1, while $|x\rangle$ transforms under the irrep D_2
2. For a tensor product state $|i\rangle \otimes |x\rangle$, rep $D(g)$'s action is

$$D(g)\,|i\rangle \otimes |x\rangle = [D_1(g)\,|i\rangle \otimes |x\rangle] + [|i\rangle \otimes D_2(g)\,|x\rangle] \tag{31}$$

For the Lie group case, the generators of different spin irreps will act independently on each kets :

$$J_a^{1\otimes 2}(|i\rangle \otimes |x\rangle) = [J_a^1\,|i\rangle] \otimes [J_a^2\,|x\rangle] \tag{32}$$

Addition of Angular Momenta

For J_3, addition is simple:

$$J_3^{j_1 \otimes j_2}(|j_1, m_1\rangle \otimes |j_2, m_2\rangle) = (m_1 + m_2)(|j_1, m_1\rangle \otimes |j_2, m_2\rangle) \tag{33}$$

but this is not the case for J_1 and J_2
The total tensor product space has the dimension $(2j_1 + 1)(2j_2 + 1)$, with states

$$|j_1, m_1\rangle \otimes |j_2, m_2\rangle \tag{34}$$

now we want to completely reduce this tensor product space into direct sum of $SU(2)$ irreps

Addition of spin $1/2$ and 1

Let's start from the highest weight state

$$\left|\frac{1}{2},\frac{1}{2}\right\rangle \otimes |1,1\rangle = \left|\frac{3}{2},\frac{3}{2}\right\rangle \tag{35}$$

Applying lowering operator $J^{-,\frac{1}{2}\otimes 1}$ to both sides :

$$J^{-}\left|\frac{3}{2},\frac{3}{2}\right\rangle = J^{-}\left|\frac{1}{2},\frac{1}{2}\right\rangle \otimes |1,1\rangle + \left|\frac{1}{2},\frac{1}{2}\right\rangle \otimes J^{-}|1,1\rangle \tag{36}$$

therefore we conclude

$$\left|\frac{3}{2},\frac{1}{2}\right\rangle = \frac{1}{\sqrt{3}}\left|\frac{1}{2},-\frac{1}{2}\right\rangle \otimes |1,1\rangle + \sqrt{\frac{2}{3}}\left|\frac{1}{2},\frac{1}{2}\right\rangle \otimes |1,0\rangle \tag{37}$$

Addition of spin $1/2$ and 1

Continuing, we arrive at this table

$$\left|\frac{3}{2},\frac{3}{2}\right\rangle = \left|\frac{1}{2},\frac{1}{2}\right\rangle \otimes |1,1\rangle \tag{38}$$

$$\left|\frac{3}{2},\frac{1}{2}\right\rangle = \frac{1}{\sqrt{3}}\left|\frac{1}{2},-\frac{1}{2}\right\rangle \otimes |1,1\rangle + \sqrt{\frac{2}{3}}\left|\frac{1}{2},\frac{1}{2}\right\rangle \otimes |1,0\rangle \tag{39}$$

$$\left|\frac{3}{2},-\frac{1}{2}\right\rangle = \sqrt{\frac{2}{3}}\left|\frac{1}{2},-\frac{1}{2}\right\rangle \otimes |1,0\rangle + \frac{1}{\sqrt{3}}\left|\frac{1}{2},\frac{1}{2}\right\rangle \otimes |1,-1\rangle \tag{40}$$

$$\left|\frac{3}{2},-\frac{3}{2}\right\rangle = \left|\frac{1}{2},-\frac{1}{2}\right\rangle \otimes |1,-1\rangle \tag{41}$$

Now these states form a subspace, transforming under spin $3/2$ irrep

Addition of spin $1/2$ and 1

We're not done yet... next we find orthogonal states

$$\left|\frac{1}{2},\frac{1}{2}\right\rangle = \sqrt{\frac{2}{3}}\left|\frac{1}{2},-\frac{1}{2}\right\rangle \otimes |1,1\rangle - \frac{1}{\sqrt{3}}\left|\frac{1}{2},\frac{1}{2}\right\rangle \otimes |1,0\rangle \tag{42}$$

$$\left|\frac{1}{2},-\frac{1}{2}\right\rangle = \frac{1}{\sqrt{3}}\left|\frac{1}{2},-\frac{1}{2}\right\rangle \otimes |1,0\rangle - \sqrt{\frac{2}{3}}\left|\frac{1}{2},\frac{1}{2}\right\rangle \otimes |1,-1\rangle \tag{43}$$

Therefore we've successfully decomposed our space into direct sums of irreps :

$$\mathbf{\frac{1}{2}} \otimes \mathbf{1} = \mathbf{\frac{3}{2}} \oplus \mathbf{\frac{1}{2}}. \tag{44}$$

(Note : The relative sign of eqn (42),(43) can be reversed)

Addition of spin $1/2$ and 1

... Really? Let's check! Denote the subspaces we've obtained as $W_{3/2}$ and $W_{1/2}$

1. Are they invariant subspaces? Yes, because for any $v \in W_{3/2}$, $J_3 v \in W_{3/2}$ and $J^{\pm} |v\rangle \subset W_{3/2}$
2. The space $W_{1/2}$ is orthogonal to $W_{3/2}$, and since generators are hermitian, $J_i W_{1/2} \in W_{1/2}$, so $W_{1/2}$ is also an invariant subspace
3. The space $W_{1/2}$ is the spin-$1/2$ rep, since the highest weight is $1/2$.

Therefore we've completely reduced our states, and the rep $\mathbf{\frac{1}{2}} \otimes \mathbf{1}$ takes the form...

$$\mathbf{\frac{1}{2}} \otimes \mathbf{1} = \begin{pmatrix} D_{3/2}(g) & 0 \\ 0 & D_{1/2}(g) \end{pmatrix} \tag{45}$$

Clebsch-Gordan Decomposition

Given $\mathbf{j} \otimes \mathbf{s}$, we may decompose it into

$$\mathbf{j} \otimes \mathbf{s} = \bigoplus_{l=|j-s|}^{j+s} \mathbf{l} \tag{46}$$

the largest 'highest weight' is $j + s$, while the smallest hightest weight' is $|j - s|$, and the coefficients C_l of the decomposition

$$|j, m\rangle \otimes |s, n\rangle = \sum_l C_l \, |l, m + n\rangle \tag{47}$$

are known as the **Clebsch-Gordan coefficients**